TSM 098 | INDEPENDENT PRACTICE

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: multiplying fractions

1. Evaluate and simplify.

$$\frac{3}{5} \cdot \left(\frac{2}{10}\right)$$

2. Evaluate and simplify.

$$\frac{-6}{8} \cdot \left(\frac{3}{2}\right)$$

3. Evaluate and simplify.

$$\frac{7}{6} \cdot \left(\frac{-2}{2}\right)$$

4. Evaluate and simplify.

$$\frac{3}{5} \cdot \left(\frac{8}{6}\right)$$

5. Evaluate and simplify.

$$\frac{-8}{6} \cdot \left(\frac{4}{3}\right)$$

6. Evaluate and simplify.

$$\frac{6}{10} \cdot \left(\frac{6}{4}\right)$$

7. Evaluate and simplify.

$$\frac{7}{3} \cdot \left(\frac{-9}{10}\right)$$

8. Evaluate and simplify.

$$\frac{-8}{6} \cdot \left(\frac{2}{3}\right)$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: dividing fractions

9. Evaluate and simplify.

$$\frac{4}{3} \div (\frac{4}{2})$$

10. Evaluate and simplify.

$$\frac{-2}{4} \div (\frac{5}{8})$$

11. Evaluate and simplify.

$$\frac{5}{5} \div (\frac{-5}{3})$$

12. Evaluate and simplify.

$$\frac{2}{3} \div (\frac{5}{8})$$

13. Evaluate and simplify.

$$\frac{-4}{12} \div (\frac{4}{6})$$

14. Evaluate and simplify.

$$\frac{4}{3} \div (\frac{1}{10})$$

15. Evaluate and simplify.

$$\frac{5}{5} \div (\frac{-3}{8})$$

16. Evaluate and simplify.

$$\frac{-6}{8} \div (\frac{8}{3})$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: mixed multiplication and division

17. Evaluate and simplify.

$$\frac{5}{8} \cdot \left(\frac{7}{3}\right)$$

18. Evaluate and simplify.

$$\frac{-3}{6} \cdot \left(\frac{2}{8}\right)$$

19. Evaluate and simplify.

$$\frac{3}{12} \cdot \left(\frac{-7}{4}\right)$$

20. Evaluate and simplify.

$$\frac{4}{3} \cdot \left(\frac{1}{6}\right)$$

21. Evaluate and simplify.

$$\frac{4}{6} \div \left(\frac{3}{3}\right)$$

22. Evaluate and simplify.

$$\frac{-2}{3} \div \left(\frac{5}{2}\right)$$

23. Evaluate and simplify.

$$\frac{6}{10} \div \left(\frac{-7}{5}\right)$$

24. Evaluate and simplify.

$$\frac{9}{12} \div \left(\frac{1}{10}\right)$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$-1 \cdot \left(\frac{5}{8}\right) \div \left(\frac{1}{2}\right)$$

26. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{6} \cdot \left(-\frac{4}{5}\right) \div \left(\frac{1}{3}\right)$$

27. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{6} \cdot \left(\frac{3}{8}\right) \div \left(-\frac{3}{2}\right)$$

28. Evaluate and simplify. Show intermediate steps.

$$-1 \cdot \left(-\frac{4}{5}\right) \div \left(-\frac{3}{4}\right)$$

29. Evaluate and simplify. Show intermediate steps.

$$-\frac{3}{4} \cdot \left(\frac{1}{3}\right) \div \left(\frac{1}{2}\right)$$

30. Evaluate and simplify. Show intermediate steps.

$$\frac{4}{3} \cdot \left(\frac{2}{5}\right) \div \left(-\frac{3}{2}\right)$$

31. Explain why dividing by a fraction is different from multiplying by that fraction.

$$\frac{2}{3} \div \frac{4}{5}$$

32. Determine whether the expression is defined. Explain.

$$\frac{3}{7} \div 0$$

TSM 098 | ANSWER KEY

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: multiplying fractions

1. Evaluate and simplify.

$$\frac{6}{50} = \frac{3}{25}$$

2. Evaluate and simplify.

$$\frac{-18}{16} = -\frac{9}{8}$$

3. Evaluate and simplify.

$$\frac{-14}{12} = -\frac{7}{6}$$

4. Evaluate and simplify.

$$\frac{24}{30} = \frac{4}{5}$$

5. Evaluate and simplify.

$$\frac{-32}{18} = -\frac{16}{9}$$

6. Evaluate and simplify.

$$\frac{36}{40} = \frac{9}{10}$$

7. Evaluate and simplify.

$$\frac{-63}{30} = -\frac{21}{10}$$

8. Evaluate and simplify.

$$\frac{-16}{18} = -\frac{8}{9}$$

TSM 098 | ANSWER KEY

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: dividing fractions

9. Evaluate and simplify.

$$\frac{4}{3} \cdot \left(\frac{2}{4}\right) = \frac{2}{3}$$

10. Evaluate and simplify.

$$\frac{-2}{4} \cdot \left(\frac{8}{5}\right) = -\frac{4}{5}$$

11. Evaluate and simplify.

$$\frac{5}{5} \cdot \left(\frac{3}{-5}\right) = -\frac{3}{5}$$

12. Evaluate and simplify.

$$\frac{2}{3} \cdot \left(\frac{8}{5}\right) = \frac{16}{15}$$

13. Evaluate and simplify.

$$\frac{-4}{12} \cdot \left(\frac{6}{4}\right) = -\frac{1}{2}$$

14. Evaluate and simplify.

$$\frac{4}{3} \cdot (10) = \frac{40}{3}$$

15. Evaluate and simplify.

$$\frac{5}{5} \cdot \left(\frac{8}{-3}\right) = -\frac{8}{3}$$

16. Evaluate and simplify.

$$\frac{-6}{8} \cdot \left(\frac{3}{8}\right) = -\frac{9}{32}$$

TSM 098 | ANSWER KEY

Multiplying and dividing numerical fractions

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: mixed multiplication and division

17. Evaluate and simplify.

$$\frac{35}{24} = \frac{35}{24}$$

18. Evaluate and simplify.

$$\frac{-6}{48} = -\frac{1}{8}$$

19. Evaluate and simplify.

$$\frac{-21}{48} = -\frac{7}{16}$$

20. Evaluate and simplify.

$$\frac{4}{18} = \frac{2}{9}$$

21. Evaluate and simplify.

$$\frac{4}{6} \cdot (\frac{3}{3}) = \frac{2}{3}$$

22. Evaluate and simplify.

$$\frac{-2}{3} \cdot (\frac{2}{5}) = -\frac{4}{15}$$

23. Evaluate and simplify.

$$\frac{6}{10} \cdot (\frac{5}{-7}) = -\frac{3}{7}$$

24. Evaluate and simplify.

$$\frac{9}{12} \cdot (10) = \frac{15}{2}$$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{4}$$

26. Evaluate and simplify. Show intermediate steps.

$$2$$

27. Evaluate and simplify. Show intermediate steps.

$$\frac{5}{24}$$

28. Evaluate and simplify. Show intermediate steps.

$$-\frac{16}{15}$$

29. Evaluate and simplify. Show intermediate steps.

$$-\frac{1}{2}$$

30. Evaluate and simplify. Show intermediate steps.

$$-\frac{16}{45}$$

31. Explain why dividing by a fraction is different from multiplying by that fraction.

$$\frac{2}{3} \cdot \frac{5}{4} = \frac{5}{6}$$

Use the reciprocal of the divisor.

32. Determine whether the expression is defined. Explain.

Undefined: no number multiplied by zero gives 3/7.